

Module 11 - Logical Operators

1. What are Logical Operators?

Logical Operators are used to combine two or more conditions and return a Boolean value (true or false).

Example Code:

I. Logical AND (&&)

→ Returns true only when both conditions are true.

```
/*  
    Cond1  Cond2  Result  
    true   true   true  
    true   false  false  
    false  true   false  
    false  false  false  
*/  
  
let age = 18;  
console.log(age >= 18 && age <= 30); // true && true = true  
  
let age1 = 35;  
console.log(age1 >= 18 && age1 <= 30); // true && false = false  
  
let age2 = 16;  
console.log(age2 >= 18 && age2 <= 30); // false && true = false // Short Circuit Evaluation  
  
let idProof = "Aadhaar";  
console.log(idProof == "PanCard" && idProof == "Driving License"); // false && false = false
```

Short-Circuit Evaluation

- It stops evaluating conditions as soon as the final result is determined.
- In **&&**, if the first condition is **false**, the remaining conditions are not evaluated.
- In **||**, if the first condition is **true**, the remaining conditions are not evaluated.

II. Logical OR (||)

→ Returns true when at least one condition is true.

```
/*  
    Cond1  Cond2  Result  
    true   true   true  
    true   false  true  
    false  true   true  
    false  false  false  
*/  
  
let pass_input = "12345";  
console.log(pass_input == "12345" || pass_input == "54321"); // true || false = true  
  
let pass_input1 = "54321";  
console.log(pass_input1 == "12345" || pass_input1 == "54321"); // false || true = true  
  
let userName = "abc";  
console.log(pass_input1 == "54321" || userName == "abc"); // true || true = true  
console.log(pass_input == "98765" || userName == "xyz"); // false || false = false
```

III. Logical NOT (!)

→ Reverses the Boolean value: true becomes false and false becomes true.

```
let num = 10 > 4;  
console.log(!num); // false  
  
let num1 = 5 > 20;  
console.log(!num1); // true  
console.log(!false); // true  
console.log(!true); // false
```